



ESPE
UNIVERSIDAD DE LAS FUERZAS ARMADAS
INNOVACIÓN PARA LA EXCELENCIA

Project Review Team 3

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1. Overview 8/10

discount and VAT included.

SUMMARY

In the commercial field, correct inventory management and price control is essential to maintain profitability and transparency in a store. Without automated monitoring, errors in price calculations, stock losses or inconsistencies in profits may occur. A system that records tire entries and exits, applies discounts and VAT automatically, and keeps the total inventory value up to date would allow managers to have exact control of stocks and profits generated, guaranteeing efficient and reliable management of the business.

BACKGROUND

In the south of Quito, specifically in the 23 de Mayo sector, tire stores play an important role in supporting local transportation and vehicle maintenance. Most of these businesses work with a variety of tire brands and sizes, serving cars, motorcycles, and trucks every day. However, many of them still rely on manual methods to record tire entries and sales, which often causes mistakes in stock control, incorrect pricing, and inconsistencies in profit calculations. In our local tire store, these issues have become evident. Sometimes, sales records do not match the actual number of tires in stock, and price updates are made inconsistently, affecting profit margins. For this reason, it is necessary to implement a simple computer-based system that can record the entry and exit of tires, automatically apply discounts and

VAT, and keep track of total inventory value. By introducing this system, the store will have more accurate control of its stock and finances, reducing human error and improving efficiency in daily operations.

A study is being carried out which shows how the progress of the years since before the pandemic where it was assumed that mobility was good and after the pandemic where it also shows the change of people for a certain way of taking care of their vehicles with the purchase of tires

When describing what the program does, it does not detail precisely and efficiently how the program will do it.

2. SRS 9/10

9. **FR09 – User Management:**

The system must allow the creation and authentication of accounts for managers and employees, including password validation.

10. **FR10 – Data Persistence:**

The system must save and load inventory and sales information using JSON files to preserve data between executions.

The requirements are detailed correctly; however, some are understood since they are a basic function performed by the program.

Features

Features for Pet shop (AGROPETCENTER)



1. Product registration

Allows you to add new products to inventory (name, category, price, quantity, tester, etc.).

2. Control of entries and exits

Registration when new products enter (purchases) or leave (sales), automatically updating the available stock.

3. Prioritized Features 9/10

3. Search and filtering of products

Make it easy to find products quickly by name, type (food, toy, medicine, etc.), or tester.

4. Automatic inventory update

Every time a sale or purchase is made, the system adjusts the amounts automatically.

The system is properly prioritized because it follows the continuous flow and execution that the system must achieve to function correctly.

4. Tasks 9/10

Task List – Pet Shop Inventory Control System

manager

1. Login as Manager

- Allows logging into the system as a manager.
- Verifies username and password (case-insensitive).
- If no account exists, it allows creating one and saves the credentials in a TXT file under the `/utils` folder.

2. Add Employee

- Creates a new employee profile with username and password.
- Stores the data in a TXT file inside `/utils/registeremployee`.
- Enables employees to log in later with their registered credentials.

3. Add Product

- Registers a new product with name, price, and initial stock quantity.
- If the product already exists, **adds the entered quantity to the existing stock**.
- Displays the updated total stock after adding.
- Products are stored in a JSON file.

4. Show Inventory

- Displays all registered products in the system.
- Shows name, price, and available stock quantity.
- If a product has **less than 4 units**, it displays a warning: *“Low stock product”*.

5. Modify Inventory

- Allows **searching a product by category or name**.
- Loads product data from the JSON file.
- Enables editing product information such as name, price, or stock quantity.

The tasks mentioned are well-directed towards the proper execution of the program; however, there are small things that can be improved by going into more detail.

5. Plan 8/10

Date

**Nov 3, 2025
(Monday)**

**Nov 4, 2025
(Tuesday)**

**Nov 5, 2025
(Wednesday)**

**Nov 6, 2025
(Thursday)**

**Nov 7, 2025
(Friday)**

**Nov 8, 2025
(Saturday)**

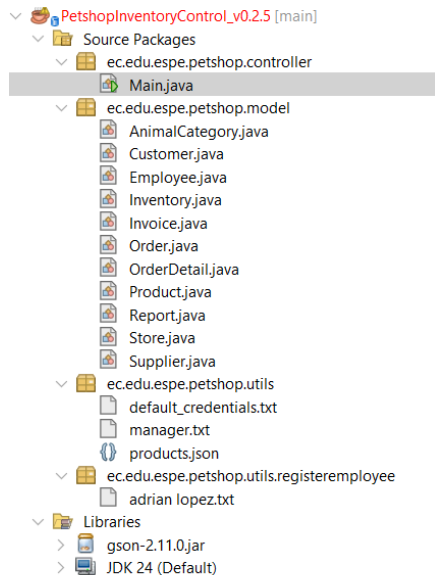
**Nov 9, 2025
(Sunday)**

**Nov 10, 2025
(Monday)**

The project organization is clearly detailed by week, but all the work is done in a single week, which overloads the teamwork.

Design

6. Architecture 9/10



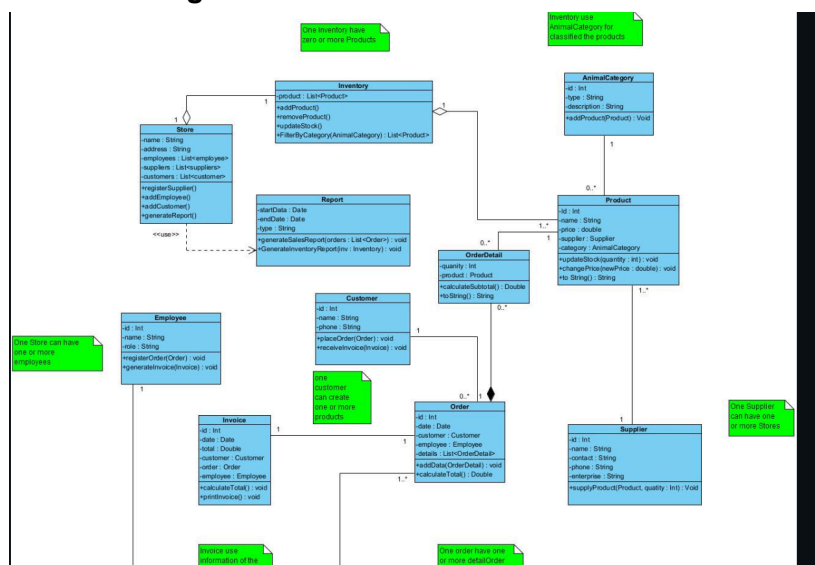
The correct way to use MVC is shown; however, it uses two forms of data persistence, which can make program coding difficult.

7. Use Case 7.5 /10



The reference image cannot be displayed correctly, which hinders the analysis of the system's use cases.

8. Class Diagram 8/10

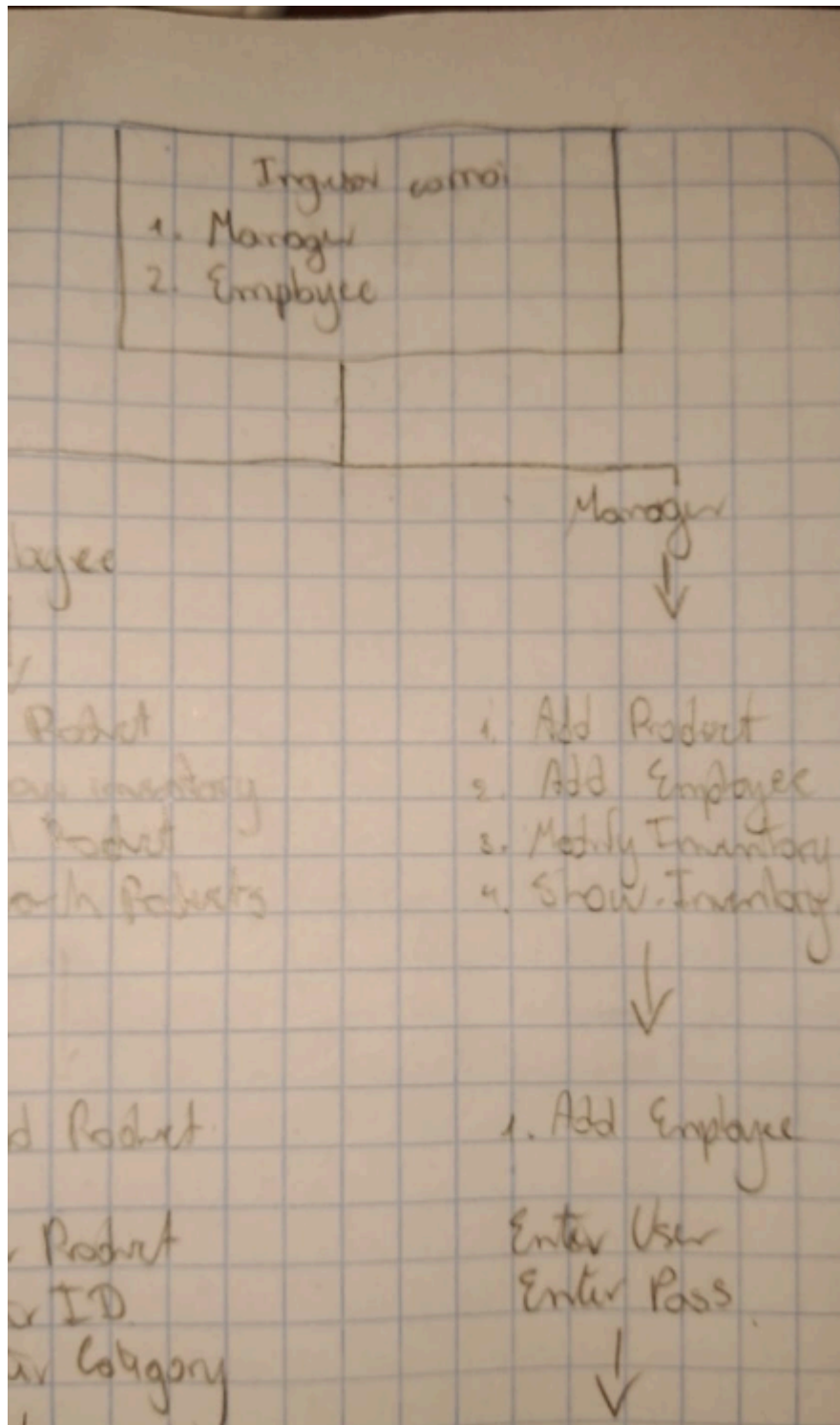


The class diagrams are well structured, but what's missing is organizing them by packages for a better understanding of each part of the program.

9. Mockups 7.5/10

PetshopInventoryControl_v0.2.5 (run) ✕

```
=== PET SHOP SYSTEM ===  
1. Login as Manager  
2. Login as Employee  
3. Exit  
Select an option: 1  
  
--- Manager Login ---  
Username: bryan  
Password: 1234  
Welcome, Manager!
```

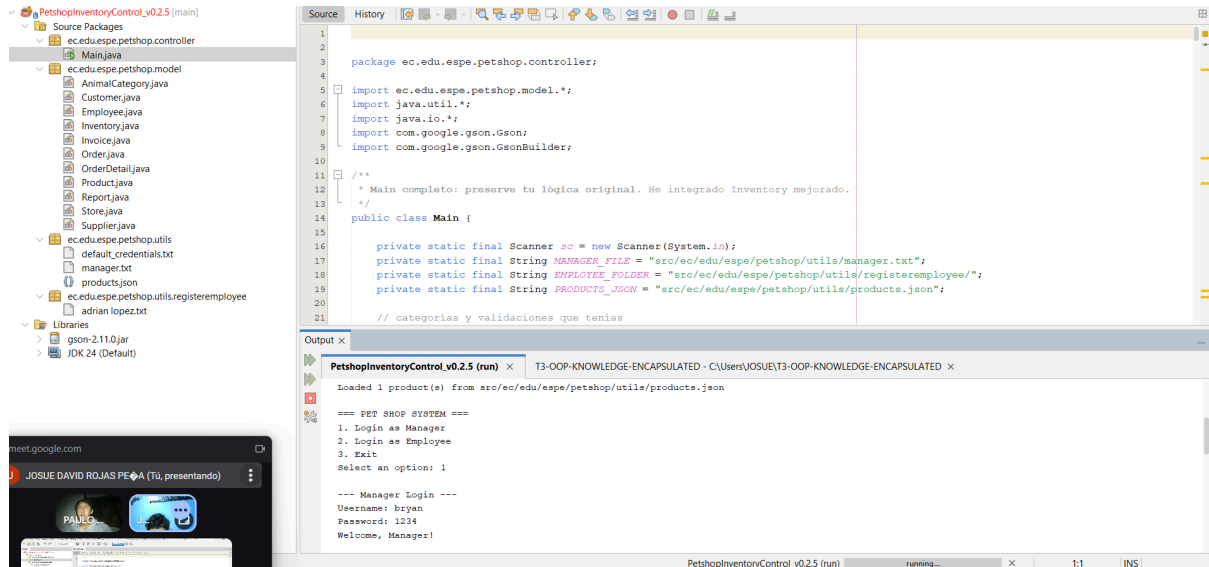


During program execution, the menus are clearly visible, but during design, we cannot distinguish the different types of menus that the project includes.

Prototype:

CMD,

10. functionality validation 10/ 10



When the program is executed, the use of validations is identified when characters are entered via the keyboard, leading to a functional and scalable program

EVIDENCE:

Ord. Artifact, evaluation/10, observations:

1. Overview 8/10
2. SRS 9/10

Features

3. Prioritized Features 9/10
4. Tasks 9/10
5. Plan 8/10

Design

6. Architecture 9/10
7. Use Case 7.5/10
8. Class Diagram 8/10
9. Mockups 7.5/10

Prototype:

- CMD,
10. functionality validation 10/10

Average 8.5/10